FORECASTING US GAS PRICES: INSIGHTS FOR CONSUMERS & BUSINESSES

TSAE Spring 2026

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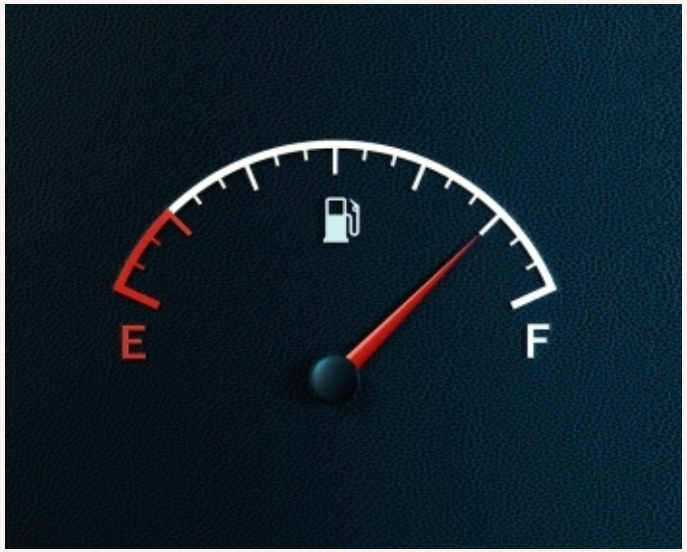

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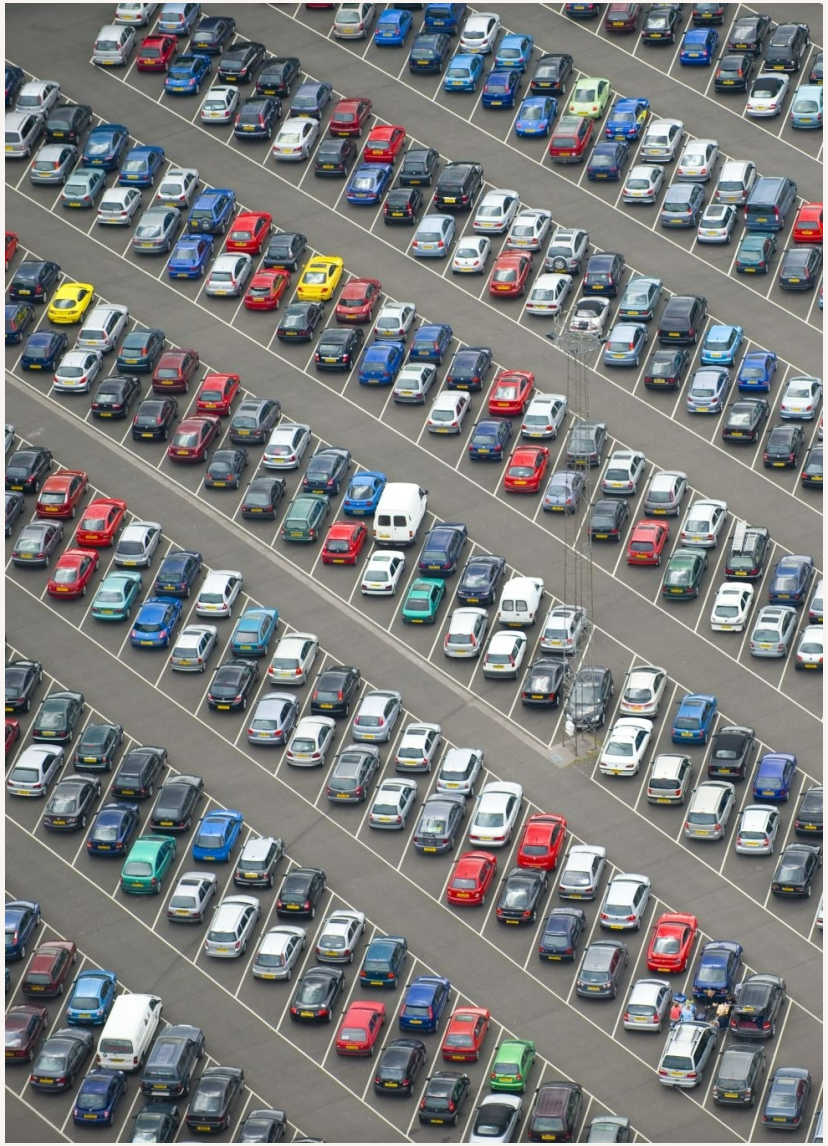
BUSINESS PROBLEM

Gasoline expenditures are a critical part of the U.S economy. It plays an important role in the budgets of businesses and consumers.

Accurate forecasts for gasoline prices enable businesses to protect their margins, and households to better plan their budgets.



DATA

Main data: Retail gasoline prices

- Monthly U.S. Gasoline Prices
- Granularity: monthly. Available since August 1990

Exogenous variables:

- Gasoline stock levels
  - https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=WGTSTUS1&f=W
  - Granularity: weekly. Available since Jan 1, 2010
- US field production of crude oil
  - https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=MCRFPUS1&f=M
  - Granularity: monthly. Available since Jan 1, 2000
- Inflation without food and energy prices
  - https://fred.stlouisfed.org/series/CORESTICKM159SFRBATL
  - Granularity: monthly or yearly. Available since 2000
- Crude oil imports
  - https://www.eia.gov/opendata/browser/petroleum/stoc/wstk
  - Granularity: monthly. Available since August 1982
  - Serves as a proxy of gasoline supply; use destination=US
- Traffic volume trends
  - https://www.fhwa.dot.gov/policyinformation/travel_monitoring/tvt.cfm
  - Granularity: monthly. Available from 1992 to 2025

DATA PREPROCESSING

Challenge	Solution
Missing data	Forward fill instead of interpolation
Level of granularity differences (monthly vs weekly)	Aggregate weekly to monthly levels – 4 week avg
Interpretability/Aggregate variables	- Total Imports (heavy sour, light sweet, medium, etc.) - Total Crude Supply (US crude oil imports + US oil production)

Major events	Dummy variables
	- COVID (March 2020 – June 2021)
	- Geopolitical conflicts
	  ○ Iraq war
	  ○ Financial crisis oil crash
	  ○ Libya war
	  ○ OPEC shale price war
	  ○ Etc.

METHODOLOGY

Pre-modeling

- EDA
- Train period: 16 years / 26 years
- Test period: 12 months (2025)

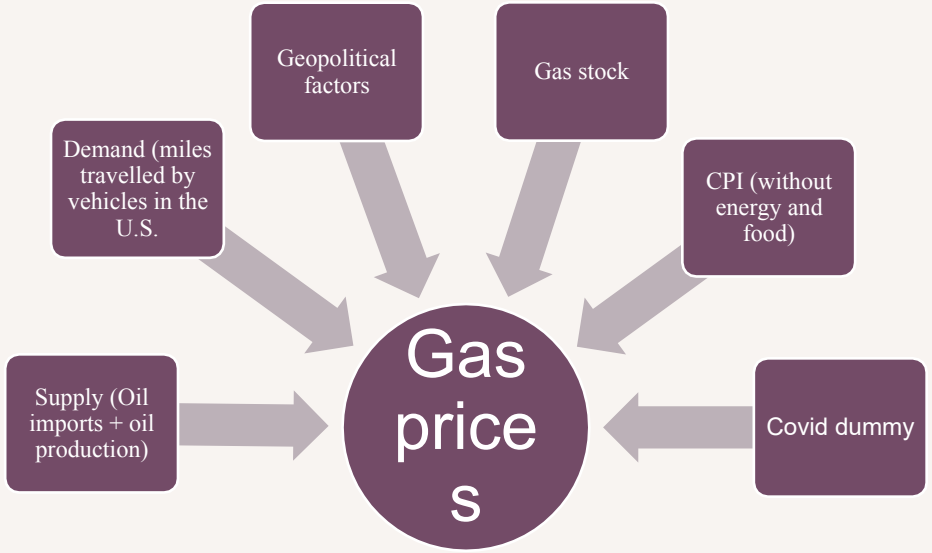
Modeling

- Three experiments
- Types of models: SARIMA, SARIMAX, Transfer Function, TBATS, and BSTS.

Post-Modeling

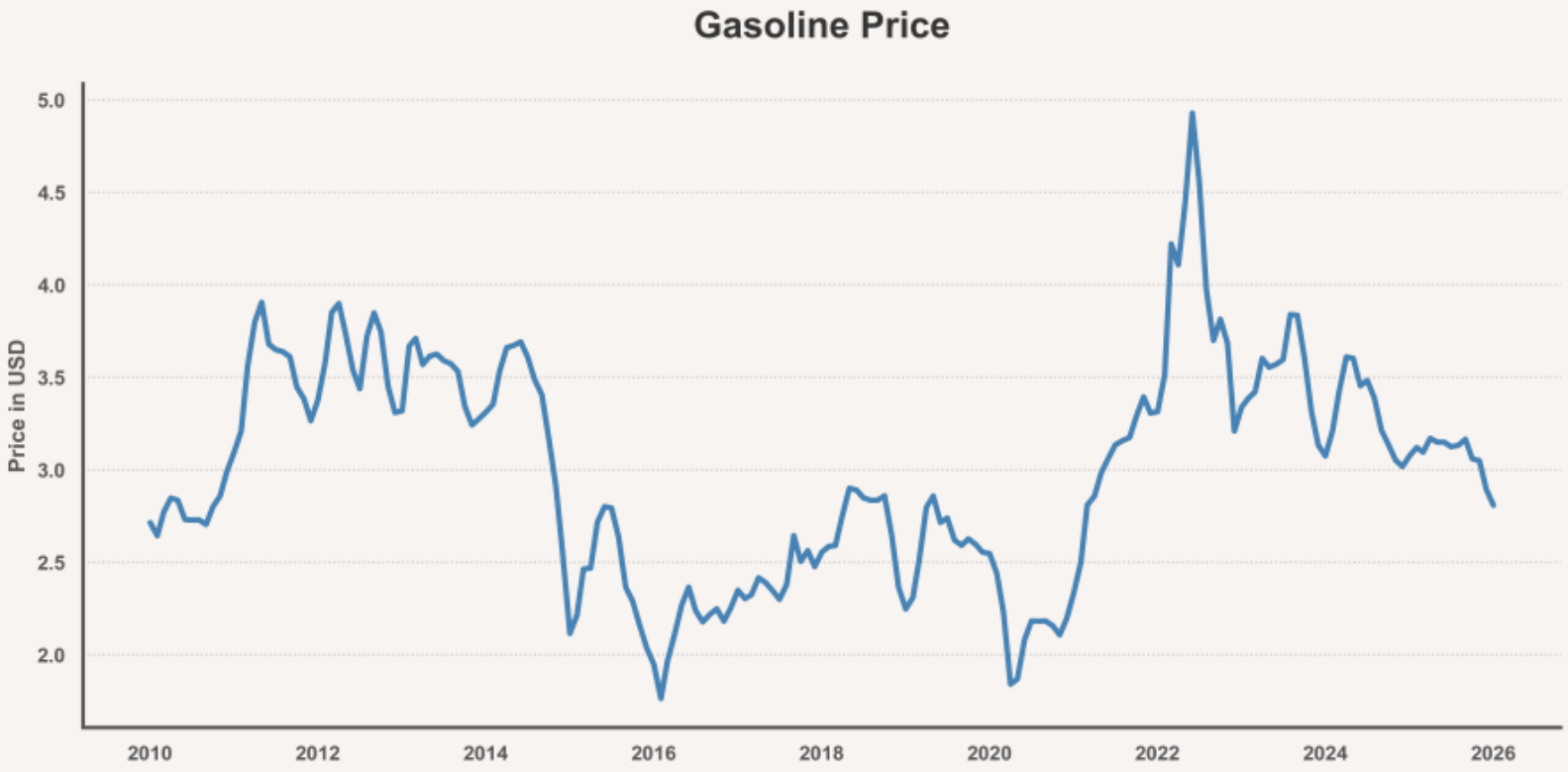
- Metrics: AIC, MAE, MAPE, and RMSE in train/test.
- Verify significance of coefficients.
- Check residuals (Correlogram, Ljung-box Test)

Target and exogenous variables

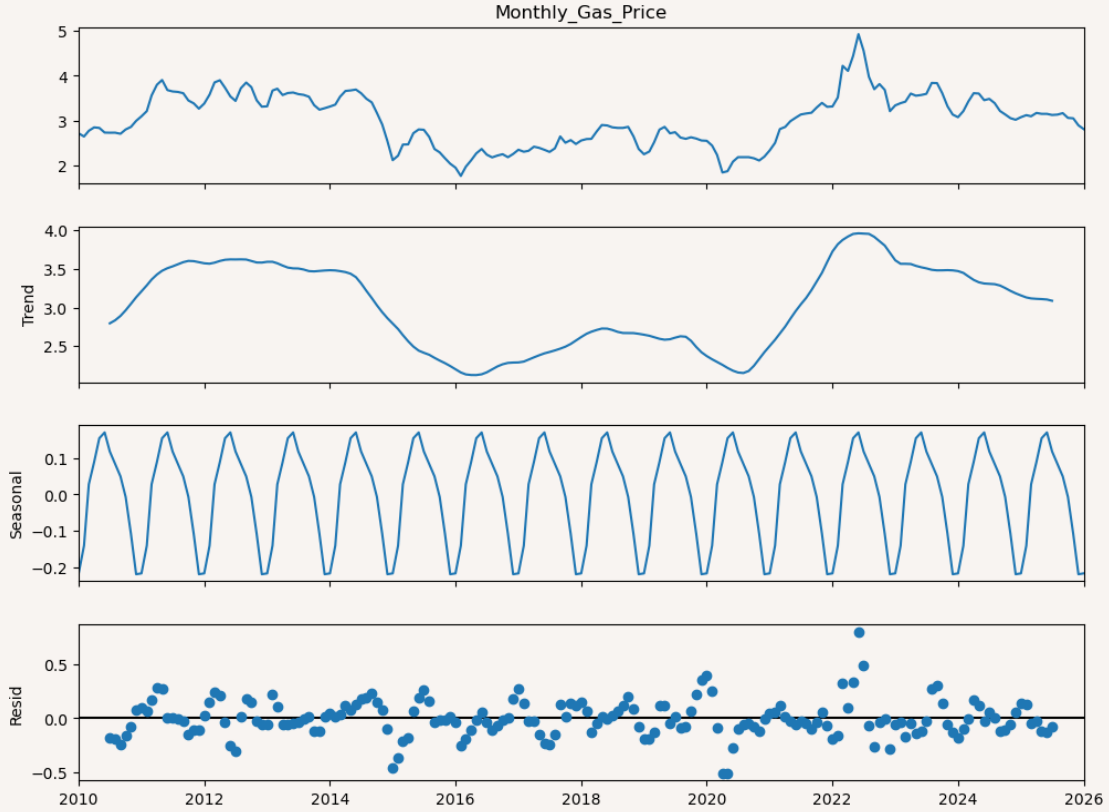


EDA

- Target variable is the average retail gasoline price in the U.S price
- The sample contains monthly data starting in Jan. 2010



EDA Seasonal decomposition



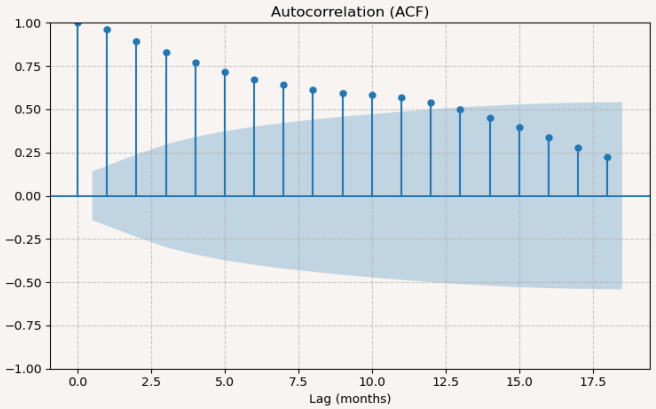
Trend: There are peaks and dips across the time series.

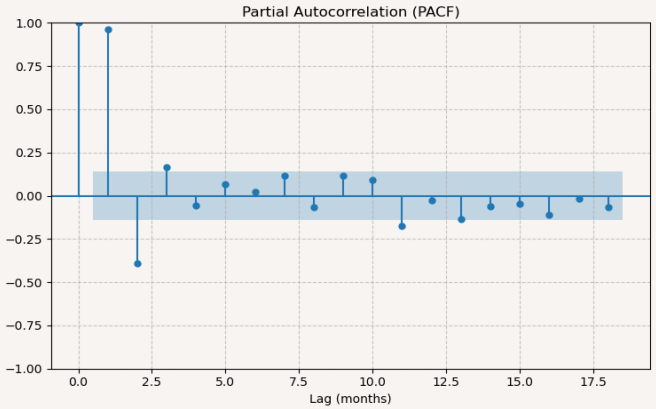
Seasonality: The graph shows a significant annual seasonality with peaks around summer.

Residuals: The residuals seem homoskedastic with some anomalies which can be captured using exogenous variables.

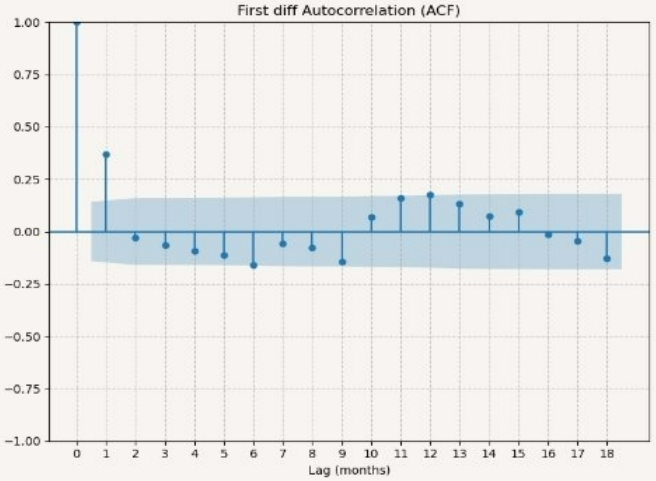
EDA Stationarity

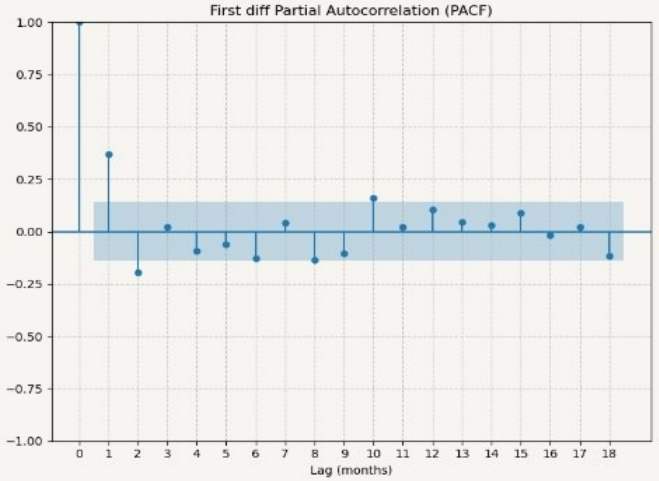
ADF test for Monthly Gas Price	
Test statistic	-2.4
p-value	0.14
Series is non-stationary	



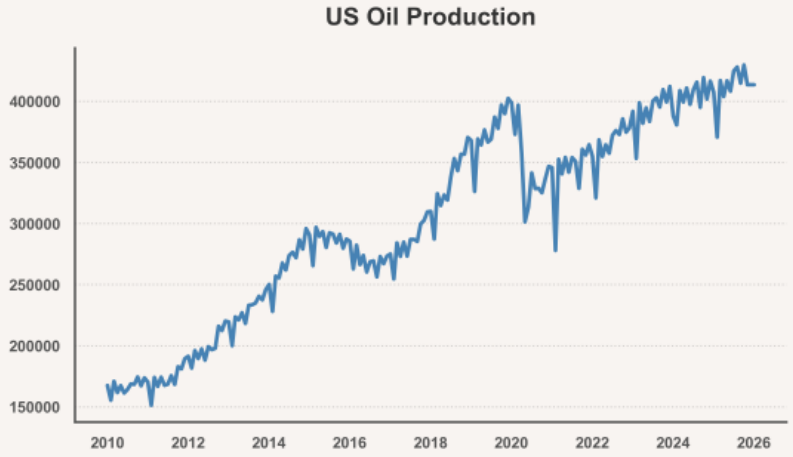


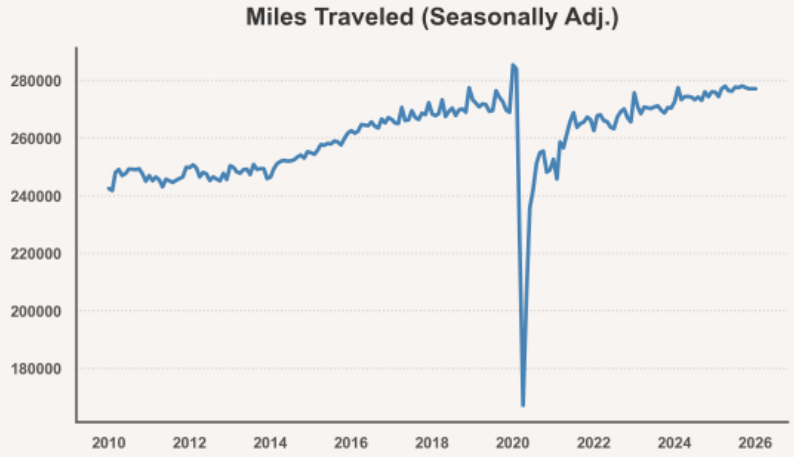
ADF test for Monthly Gas Price (first difference)	
Test statistic	-4.65
p-value	0.0001
Series is stationary	

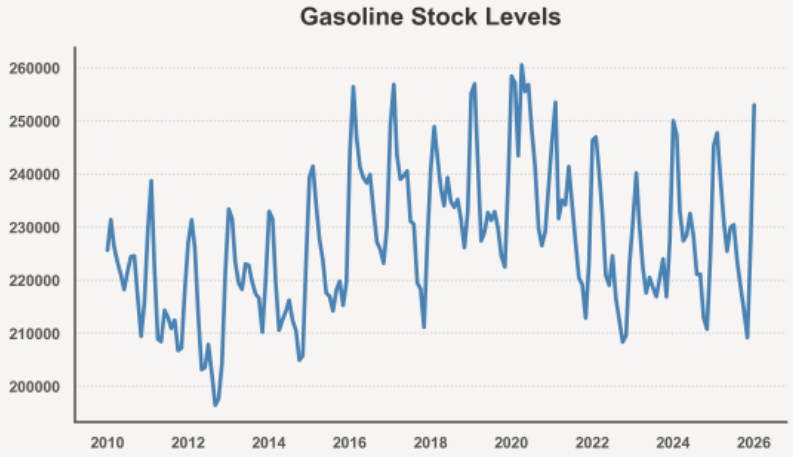


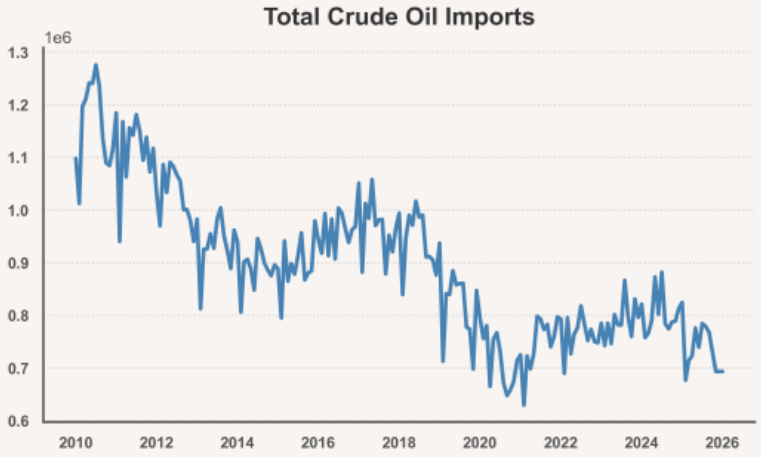


EDA Exogenous variables









EDA Granger Causality Test

$H_0 =$ The time series X does not Granger cause the time series Y.
$H_a =$ The time series X has a Granger cause of the time series Y.

Target: Monthly Gasoline price

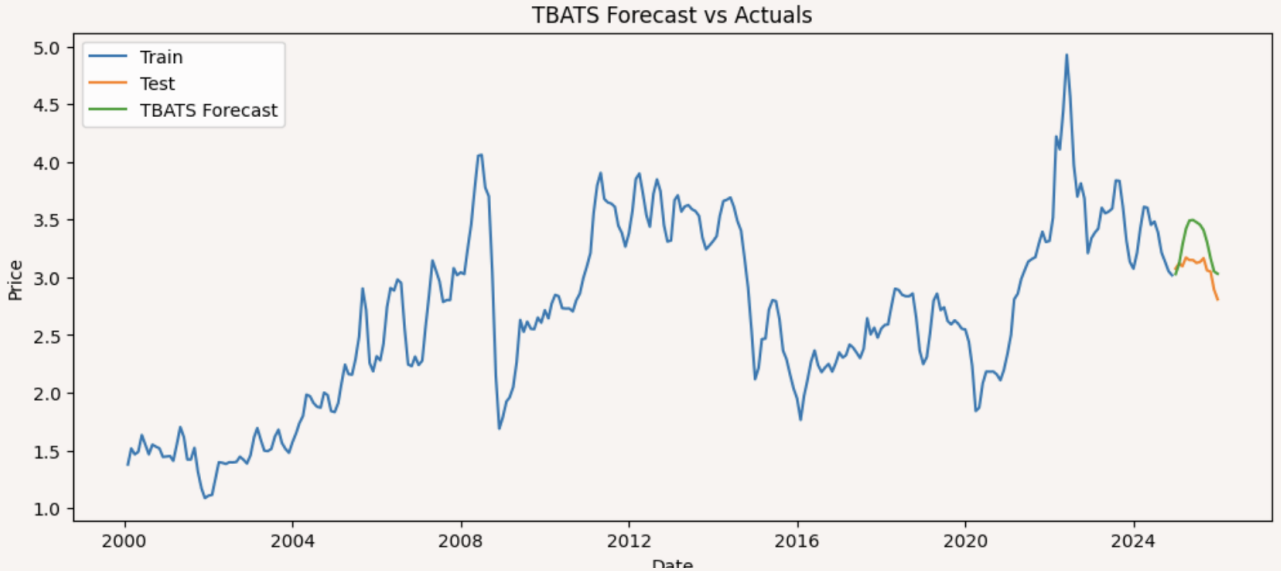
Exogenous variables	Granger Causality
Gasoline Stock	Yes
Oil production	Yes
Miles traveled sa	Yes
Covid dummy	Yes
Geopolitical dummy	Yes

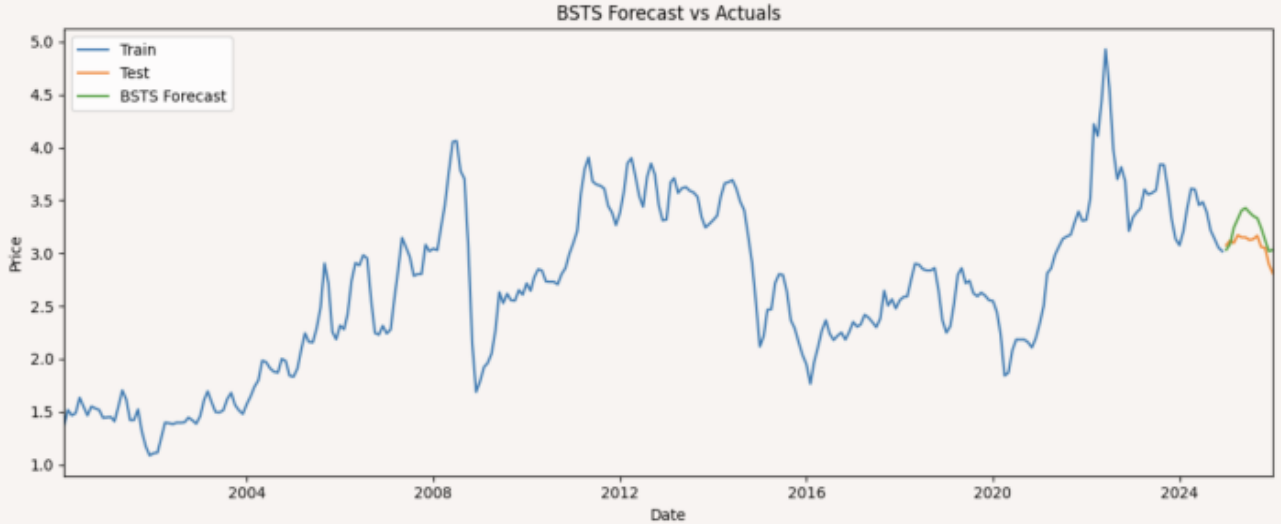
Exogenous variables	Granger Causality
Core Sticky CPI	No
Import heavy sweet	No
Import heavy sour	No
Import light sour	No
Import light sweet	No
Import medium	No
Total imports	No
Total crude supply	No
Miles traveled nsa	No

FIRST EXPERIMENT

- 2000-2026 years
  - Differenced target variable and non-stationary exogenous variables per ADF
  - Selected exogenous variables per Gran Causality
- Models:
  - TBATS
  - BSTS; no significant exogenous variables :(

Model	MAE	MAPE	RMSE
TBATS	0.22	7.05%	0.24
BSTS	0.17	5.39%	0.18





SECOND EXPERIMENT

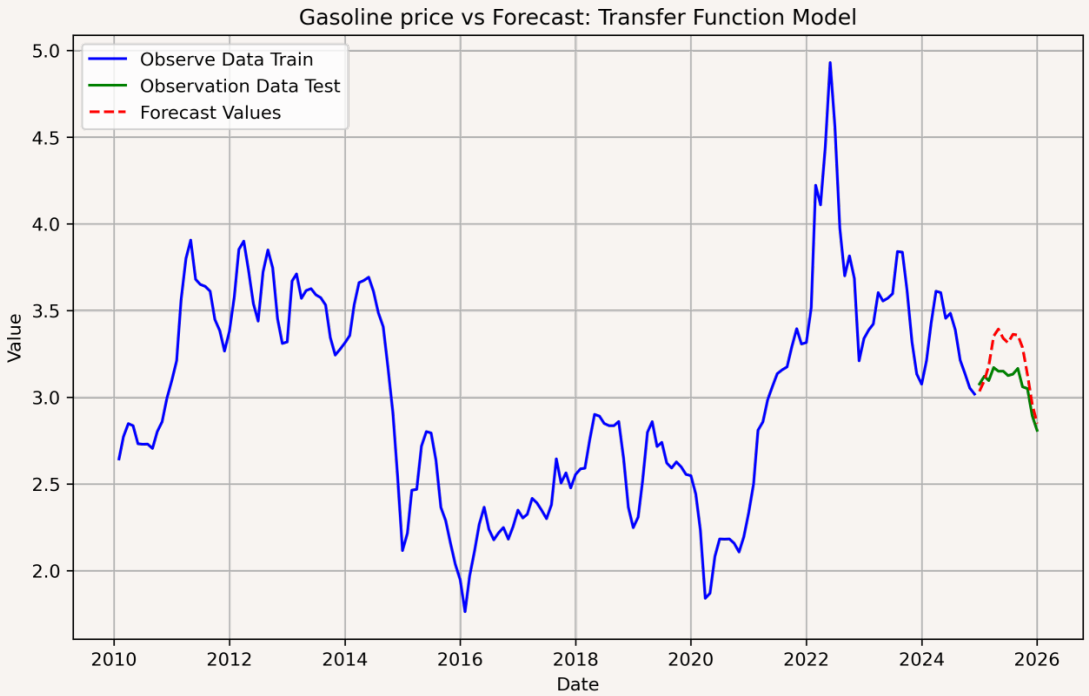
Key decisions:

- Apply the first difference of the Gasoline price.
- Try exogenous factors that pass the Granger Causality.
- Use ACF/PACF to establish non-seasonal and seasonal components.

Try the SARIMA Family Models

Model	MAE	MAPE	RMSE
SARIMA	0.28	8.43%	0.26
SARIMAX	0.19	5.42%	0.17
Transfer Function (SARIMAX)	0.16	4.43%	0.14

Exogenous variables: Covid, Gasoline Stock and, lag 1 of Gasoline stock.

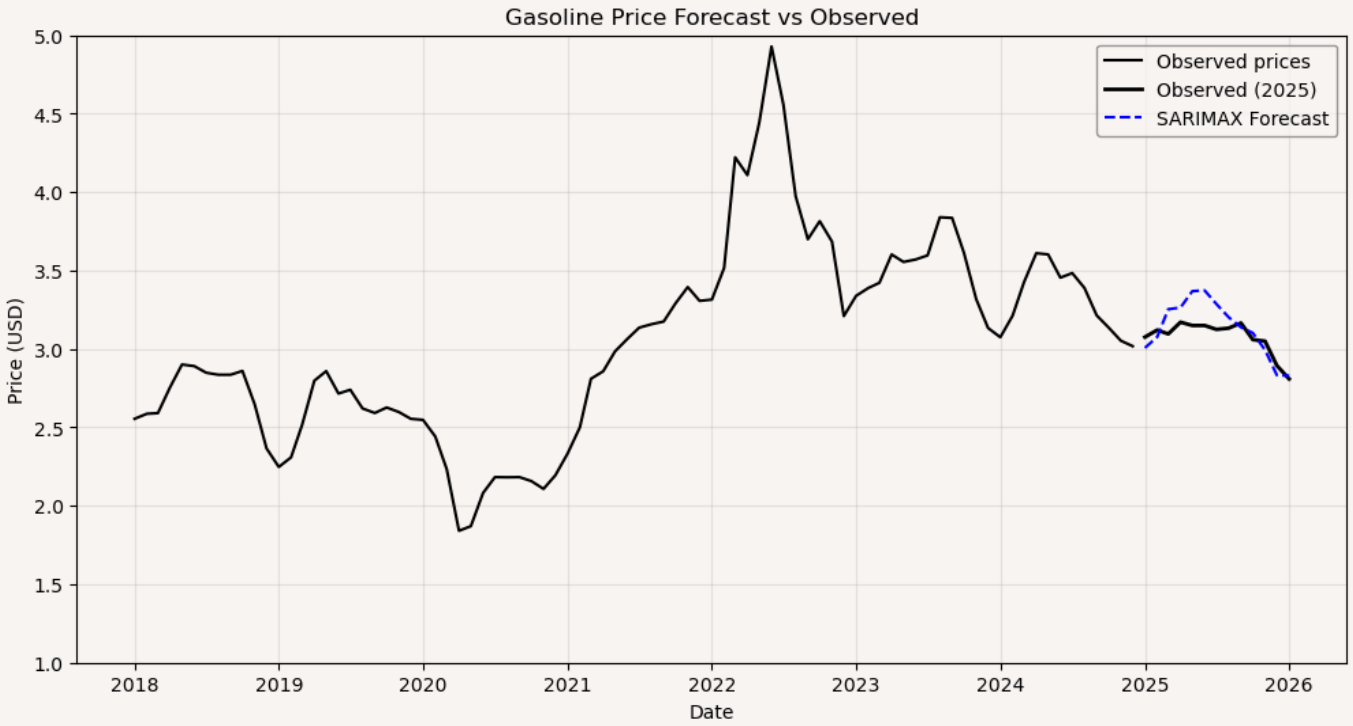


Forecast transfer function (0, 1, 1)x(1, 1, 0, 12)

THIRD EXPERIMENT

- Lagging features to use the model for prediction.
- **Exogenous variables:** Covid, gasoline stock levels, miles traveled, crude oil supply and geopolitical dummy.
- SARIMAX(2, 0, 0)x(2, 1, 1)$_{12}$

Model	MAE	MAPE	RMSE
Log-differencing + SARIMAX	0.095	3.08%	0.013



RECOMMENDATIONS AND FUTURE MODELS

- **Regional Forecasting:** We model U.S. national average gasoline price, but future iterations model prices at the state or regional level. To do this effectively, the model's feature set must be updated to include regional data.
- **NLP Analysis**: Social media increasingly reflects geopolitical events and public sentiment surrounding energy markets. Future work could apply NLP techniques to social media data to extract sentiment or event-based signals. These signals could be incorporated as exogenous variables in forecasting models to potentially improve gasoline price predictions.
- **Key consideration:** For future forecasting, it is highly recommended to apply a log transformation to the target variable to handle the trend and heteroskedasticity in gasoline prices.

THANK YOU

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